

AI-Generated Image Detection: Cross-Generator Generalization & Unsupervised Clustering

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4. Experimental Results

4.1 Main Results: Accuracy (%) on All Test Sets

Method	Backbone	Strategy	GLIDE (in-dist)	BigGAN (cross)	ADM (cross)	VQDM (cross)
ResNet-18	~11 M	Full FT	98.21	99.30	56.20	53.00
ViT-Small	~22 M	Full FT	98.70	97.80	54.90	52.10
ResNet-18	~11 M	Linear Probe	92.03	91.90	59.40	54.70
CLIP-ViT	~86 M	Linear Probe	99.30	98.30	61.30	55.60
CLIP-ViT	~86 M	KNN (k=1)	85.40	81.70	64.90	64.70
ResNet-18	~11 M	Patch 3×3	61.45	54.00	60.80	61.80

Bold red: best per column.

1. Overview

- Task:** Distinguish AI-generated images (diffusion / GAN) from authentic photographs.
- Dataset:** GenImage subset – GLIDE (train source), BigGAN / ADM / VQDM (cross-generator evaluation)
- Models:** ResNet-18, ViT-Small, CLIP-ViT with multiple training strategies
- Environment:** Python 3.12, PyTorch 2.6.0+cu124, RTX 4060 Laptop GPU

2. Dataset Preparation

2.1 Generators Downloaded

Generator	Size	Images	Role
GLIDE	1.2 GB	~12,000	Train
BigGAN	934 MB	~12,000	Cross
ADM	1.5 GB	~12,000	Cross
VQDM	1.4 GB	~12,000	Cross

2.2 Subset Split

Subset	Source	Real	Fake	Total
train	GLIDE	4,000	4,000	8,000
val	GLIDE	1,000	1,000	2,000
test/glide	GLIDE	~502	500	~1,002
test/biggan	BigGAN	500	500	1,000
test/adm	ADM	500	500	1,000
test/vqdm	VQDM	500	500	1,000

3. Methods

3.1 Architectures

Backbone	Parameters	Pretrained
ResNet-18	~11 M	ImageNet-1K
ViT-Small (patch16_224)	~22 M	ImageNet-21K (timm)
CLIP-ViT-B/32	~56 M	OpenAI CLIP (400M)

3.2 Training Strategies

- Full Finetune (FT)**
Update all backbone + classifier parameters end-to-end.
- Linear Probe (LP)**
Freeze backbone; train only a single linear classification head.
- KNN (k=1) (Ojha et al.)**
Freeze backbone; classify by cosine nearest-neighbour against train memory bank.

3.3 Hyperparameters

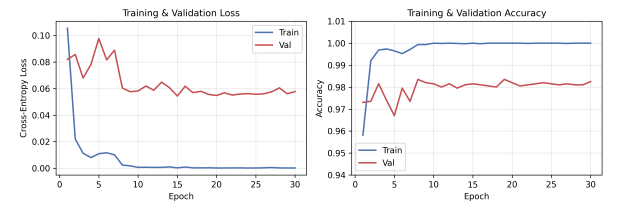
Hyperparameter	Full FT	Linear Probe
Epochs	30	30
Batch size	64	32 (CLIP), 64 (others)
Optimizer	AdamW	AdamW

4.2 Full Metrics – Baseline (ResNet-18 Full FT)

Test Set	Acc (%)	Prec (%)	Rec (%)	F1	AUC
GLIDE (in-dist)	98.21	97.83	98.61	98.22	99.8
BigGAN (cross)	99.30	100.00	98.60	99.30	99.8

Bold red: best per column.

4.3 Training Curves (ResNet-18 Full FT)



- Train loss converges to ~0 after epoch 5
- Val accuracy plateaus at ~98.5% from epoch 9
- Mild overfitting: train acc ~100% vs val acc ~98.5%
- Best Val Acc: 98.55% (epoch 23)**

4.4 Patch-Level Analysis (ResNet-18, 3×3 Grid)

- Tested with 3×3 patch grid
- Each patch classified independently; final prediction by majority vote
- Mean consistency = fraction of patches agreeing with the majority vote

Test Set	Accuracy (%) (majority vote)	Mean Consistency (%) (patch agreement)
GLIDE (in-dist)	61.45	92.40
BigGAN (cross)	54.00	86.54
ADM (cross)	60.80	89.62
VQDM (cross)	61.80	89.62

4.5 Unsupervised Generator Clustering

(200 fake images per generator)

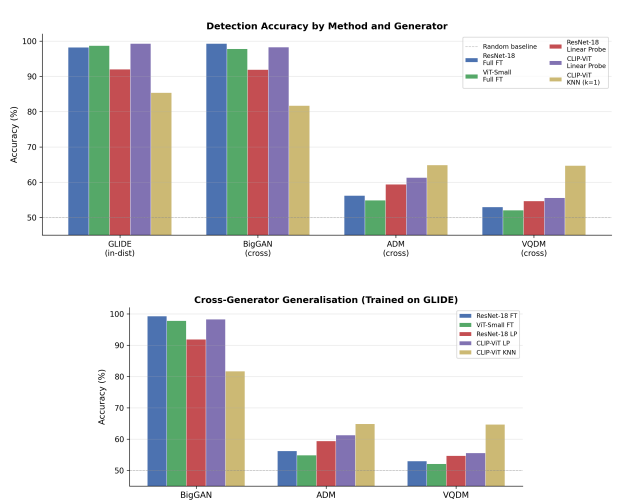


Fine-tuned Cluster Breakdown

Cluster	GLIDE	BigGAN	ADM	VQDM	Dominant
0	192	103	14	9	GLIDE
1	0	0	75	79	Diffusion
2	8	97	40	33	BigGAN
3	0	0	71	79	Diffusion

- Cluster 0 → dominated by GLIDE
- Clusters 1 + 3 → ADM/VQDM (diffusion family)
- Cluster 2 → BigGAN
- Task-oriented fine-tuning implicitly separates generator families without any generator-identity supervision.

4.6 Cross-Generalization Summary



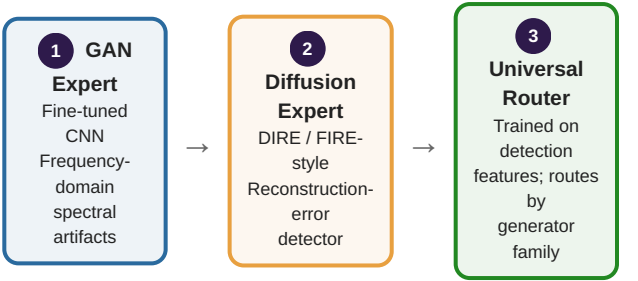
Chance = 50%

5. Key Findings

- ✓ **1. Full Finetune Dominates In-Distribution and GAN**
ResNet-18 and ViT-Small FT achieve ~98–99% on GLIDE and BigGAN, confirming strong generator-specific artifact learning.
- ✗ **2. All Methods Fail on ADM and VQDM**
All detectors drop to 52–65% on ADM and VQDM — barely above random (50%), suggesting fundamentally different artifacts.
- ✓ **3. CLIP KNN Shows Better Generalization**
CLIP-ViT KNN achieves 64.9%/64.7% on ADM/VQDM — the best of any method, supporting frozen vision-language features.
- ✓ **4. Linear Probe > KNN on Easy Generators**
CLIP-ViT Linear Probe (99.3%/98.3%) outperforms KNN (85.4%/81.7%) with a learned linear boundary.
- ✓ **5. High Patch Consistency Despite Low Accuracy**
~61% accuracy but ~90% consistency — patches agree even when wrong. Errors are systematic.
- ✓ **6. Fine-tuned Features Encode Generator Identity**
K-Means on fine-tuned ViT-Small (Purity=0.559, NMI=0.378) outperforms pretrained (0.278/0.003) by 2×.

MoE Framework (Future Directions)

Three-component architecture validated by Findings 5 & 6:



Why This Matters

Detection-oriented supervision naturally organizes the feature space by generator family. This makes it possible to build a lightweight router that dispatches test images to specialized experts (GAN / Diffusion) without needing explicit generator labels.

Figures (see code / outputs / ablation / figures)

